

SULIT



SECP2523 DATABASE

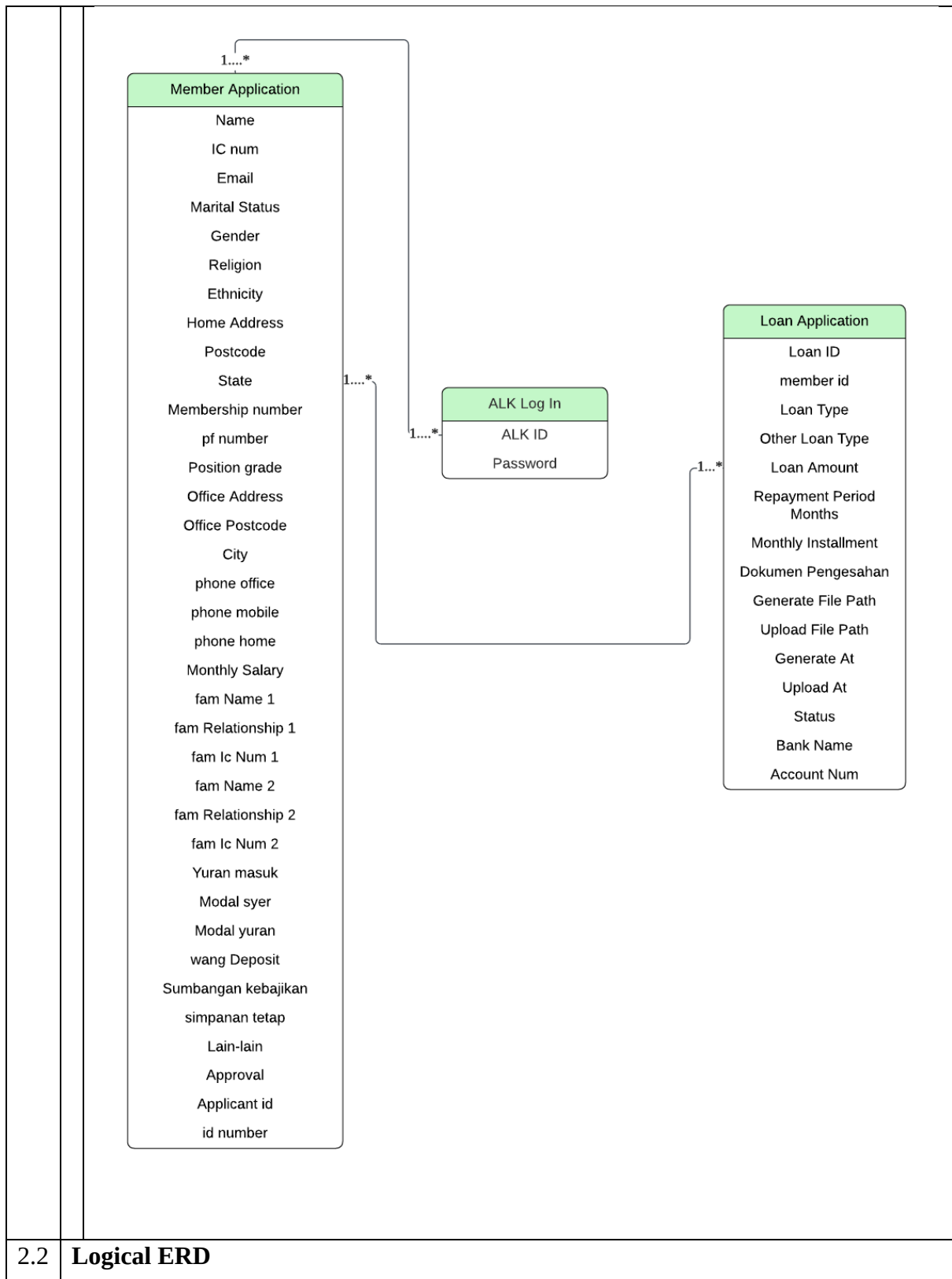
SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

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IC No. / Matric No.	040302-03-0608 / A23CS0220
Year / Program	2 / SECPH
Section	02
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Case Study	ALK module
System Name	Koperasi KADA Membership & Loan Application System
Group Name	CO-T

SECTION A (SYSTEM’S OVERVIEW)															
1.1	Overall Description														
	Overview proposed system: Koperasi KADA realizes the importance of technology and wants to improve its membership and loan application processes to better serve its members. Currently, the organization faces challenges due to manual and unorganized processes, leading to inefficiencies, inconsistent data, and slow services. These issues make it harder to provide timely assistance and benefits, which negatively affects member satisfaction and engagement. To address these problems, it is recommended to implement an advanced membership and loan management system. This system would automate and streamline processes such as member registration, loan application processing, and payment tracking, all within a centralized platform. By integrating these functions, the system ensures that member and loan data is accurate, secure, and easy to access. Automating repetitive tasks will also allow the team to focus on more important activities like financial planning and improving member engagement.														
		<table><tr><th>Group Members’ Name</th><th>Matric No</th><th>Module</th></tr><tr><td>Nurul Ika Syafiny binti Azhar</td><td>A23CS0164</td><td>Staff/Admin</td></tr><tr><td>Nuraisyah binti Mohd Zikre</td><td>A23CS0160</td><td>Non-member</td></tr><tr><td>Harini A/P Sangaran</td><td>A23CS0081</td><td>Member</td></tr></table>	Group Members’ Name	Matric No	Module	Nurul Ika Syafiny binti Azhar	A23CS0164	Staff/Admin	Nuraisyah binti Mohd Zikre	A23CS0160	Non-member	Harini A/P Sangaran	A23CS0081	Member	
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1.2	Project Weaknesses or Improvement														
	ALK module weaknesses:														

		• ALK members handle a large volume of physical documents when approving membership and loan applications, which is inefficient and prone to errors. • Every application requires a physical meeting of ALK members for review and approval, causing delays and dependency on scheduling. • The list of applicants is maintained manually on paper, making it difficult to organize, update, or retrieve information quickly. This increases the risk of data loss and errors.
1.3	Proposed Module	
		ALK module system improvements: • Allow applicants to submit their documents online and enable ALK members to review and process them digitally, removing the need for physical paperwork. • Introduce an automated approval process where ALK members can review applications online and provide approvals without needing frequent in-person meetings. • Replace the paper-based applicant list with a centralized digital database. This database should allow ALK members to search, filter, and update applicant details efficiently while maintaining data accuracy and security. • Add features for real-time status updates and notifications, ensuring that ALK members stay informed about pending applications and can act promptly. • Ensure all records, including applicant lists and approval statuses, are securely stored within the system to reduce the risk of data loss and improve long-term accessibility.
SECTION B (DATABASE PLANNING AND DESIGN)		
2.1	Conceptual ERD	



	Name	VARCHAR	Full name of the applicant.
	IC_num	VARCHAR	Identification card number.
	Email	VARCHAR	Applicant's email address.
	Marital_Status	ENUM NOT NULL	Marital status of the applicant ('Bujang', 'Berkahwin', 'Duda' & 'Janda')
	Gender	ENUM NOT NULL	Gender of the applicant. ('Lelaki' & 'Perempuan')
	Religion	VARCHAR	Religion of the applicant.
	Ethnicity	VARCHAR	Ethnicity of the applicant.
	Home_Address	TEXT	Home address of the applicant.
	Postcode	VARCHAR	Postal code of the applicant's home.
	State	VARCHAR	State of residence of the applicant.
	Membership_number	VARCHAR	Unique membership number assigned.
	pf_number	VARCHAR	Provident fund number.
	Position_grade	VARCHAR	Grade of the applicant's job position.
	Office_Address	TEXT	Address of the applicant's workplace.
	Office_Postcode	VARCHAR	Postal code of the workplace.
	City	VARCHAR	City of residence of the applicant.
	phone_office	VARCHAR	Office phone number of the applicant.
	phone_mobile	VARCHAR	Mobile phone number of the applicant.
	phone_home	VARCHAR	Home phone number of the applicant.
	Monthly_Salary	DECIMAL	Monthly salary of the applicant.
	famName1	VARCHAR	Name of the first family member.

famRelationship1	VARCHAR	Relationship to the first family member.
famIcNum1	VARCHAR	IC number of the first family member.
famName2	VARCHAR	Name of the second family member.
famRelationship2	VARCHAR	Relationship to the second family member.
famIcNum2	VARCHAR	IC number of the second family member.
Yuran_masuk_value	DECIMAL	Entry fee for membership.
Modal_syer_value	DECIMAL	Share capital value.
Modal_yuran_value	DECIMAL	Subscription value.
wang_Deposit_value	DECIMAL	Deposit amount.
Sumbangan_kebajikan_value	DECIMAL	Contribution to welfare.
simpanan_tetap_value	DECIMAL	Fixed savings amount.
Lain_lain_value	DECIMAL	Other fees.
Approval	ENUM	Indicates whether the application is pending, reviewed, approved or disapproved.
Applicant_id (PK)	INT	Primary key, unique identifier for each record.
id_number	VARCHAR	Unique ID number of the applicant.

Table: ALK

Column Name	Data Type	Description
ALK_ID (PK)	VARCHAR	Primary key, unique identifier for ALK.
Password	VARCHAR	Password for ALK login.

Table: Loan_Application

Column Name	Data Type	Description
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	Loan_ID (PK)	INT	Primary key, unique identifier for the loan.
	member_id (FK)	INT	Foreign key referencing Applicant_id.
	LoanType	VARCHAR	Type of loan applied for.
	OtherLoanType	VARCHAR	Other loan type, if applicable.
	LoanAmount	DECIMAL	Amount of the loan requested.
	RepaymentPeriodMonths	INT	Loan repayment period in months.
	MonthlyInstallment	DECIMAL	Monthly repayment amount.
	Dokumen_Pengesahan	BOOLEAN	Verification document details.
	GenerateFilePath	VARCHAR	File path for generated documents.
	UploadFilePath	VARCHAR	File path for uploaded documents.
	GenerateAt	DATETIME	Timestamp for document generation.
	UploadAt	DATETIME	Timestamp for document upload.
	Status	VARCHAR	Current status of the loan application.
	BankName	VARCHAR	Name of the bank for repayment.
	Account_Num	DECIMAL	Bank account number for repayment.
2.4	Relational Database Schema		
	MEMBER_APPLICATION(Name, IC_Num, Email, Marital_Status, Gender, Religion, Ethnicity, Home_Address, Postcode, State, membership_number, pf_number, position_grade, office_address, office_postcode, city, phone_office, phone_mobile, phone_home, monthly_salary, famName1, famRelationship1, famIcNum1, famName2, famRelationship2, famIcNum2, yuran_masuk_value, modal_syer_value, modal_yuran_value, wang_deposit_value, sumbangan_kebajikan_value, simpanan_tetap_value, lain_lain_value, approval, applicant_id, id_number) LOAN_APPLICATION(Loan_ID, member_id, LoanType, OtherLoanType, LoanAmount, RepaymentPeriodMonths, MonthlyInstallment,		

	Dokumen_Pengesahan, GenerateFilePath, UploadFilePath, GenerateAt, UploadAt, Status, BankName, Account_Num) ALK(ALK_id, password)
2.5	Normalization
	1. Functional Dependencies (FD) (MEMBER_APPLICATION) - FD1: applicant_id $\rightarrow$ name, IC_Num, email, marital_status, gender, religion, ethnicity, home_address, postcode, state, membership_number, pf_number, position_grade, office_address, office_postcode, city, phone_office, phone_mobile, phone_home, monthly_salary - FD2: applicant_id $\rightarrow$ famName1, famRelationship1, famIcNum1, famName2, famRelationship2, famIcNum2 - FD3: applicant_id $\rightarrow$ yuran_masuk_value, modal_syer_value, modal_yuran_value, wang_deposit_value, sumbangan_kebajikan_value, simpanan_tetap_value, lain_lain_value (LOAN_APPLICATION) - FD4: Loan_ID $\rightarrow$ member_id, LoanType, OtherLoanType, LoanAmount, RepaymentPeriodMonths, MonthlyInstallment, Dokumen_Pengesahan, GenerateFilePath, UploadFilePath, GenerateAt, UploadAt, Status, BankName, Account_Num - FD5: member_id $\rightarrow$ Loan_ID (assumed foreign key relation) (ALK) - FD6: ALK_id $\rightarrow$ password

2. 2NF RELATION

R1: USER (applicant information)

- Primary Key: applicant_id

R2: FAMILY_INFORMATION (family details related to the user)

- Primary Key: applicant_id

R3: CONTACT_DETAILS (user's contact information)

- Primary Key: applicant_id

R4: MEMBER_PROFILE (user's financial profile)

- Primary Key: applicant_id

R5: LOAN_APPLICATION (loan application information)

- Primary Key: Loan_ID

R6: ALK (user login credentials)

- Primary Key: ALK_id

3. 3NF RELATION

All relations are already in 3NF as they have no transitive dependencies

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